

Multiple Choice (10 marks)

1. Message authentication is a service beyond
 - (a) Message Confidentiality
 - (b) Message Integrity
 - (c) Message Splashing
 - (d) Message Sending
2. A sender must not be able to deny sending a message that was sent, is known as
 - (a) Message Nonrepudiation
 - (b) Message Integrity
 - (c) Message Confidentiality
 - (d) Message Sending
3. Which of the following is an authentication method?
 - (a) Secret question
 - (b) Biometric
 - (c) SMS code
 - (d) All of the above
4. Public key encryption is advantageous over Symmetric key Cryptography because of
 - (a) Speed
 - (b) Space
 - (c) Key exchange
 - (d) Key length
5. Which of the following is true of mutation-based fuzzing?
 - (a) It generates each different input by modifying a prior input
 - (b) It works by making small mutations to the target program to induce faults
 - (c) Each input is mutation that follows a given grammar
 - (d) It only makes sense for file-based fuzzing, not network-based fuzzing

True / False (10 marks)

Answer True or False

6. True or False: Encryption and decryption do not ensure data integrity, which is the assurance that information remains unaltered during transmission.
7. True or False: The Silk Road was a notorious Dark Web marketplace known for the sale of illegal drugs and other illicit goods.
8. True or False: A digital signature can be effectively implemented using a shared-key system, allowing both parties to sign and verify the data.
9. True or False: The message must be encrypted at the sender site and decrypted at the receiver site to ensure secure communication.
10. True or False: Wireshark is a widely used tool for wireless traffic sniffing and network protocol analysis.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to check whether the entered number is greater than the elements of the given array.
12. Write a function to check if a nested list is a subset of another nested list.

End of Paper